

The T0 Energy Field Model: Mathematical Formulation

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Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- E_0 : The two values are the same E_0 , bare ($\sqrt{m_e m_\mu} = 7.348$ MeV) and fractally corrected ($E_0/\sqrt{K_{\text{frak}}} = 7.398$ MeV, Doc. 190, R72). The corrected value serves as the anchor for α_i from the masses alone, $\alpha = 1/137.059$ (+0.017 %).
- α : agreement 0.0005 % instead of 0.03 %.
- Anomalous magnetic moments: the short formula $a_\ell = \frac{\xi}{2\pi}(E_\ell/E_e)^2$ has been removed; recalculated it gave $a_\mu = 0.907$ instead of 1.166×10^{-3} . Authoritative are Docs. 018 and 158.
- Dimension of the Lagrangian density: $[E^4]$, not $[E^2]$.
- G_{SI} formula: $\frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} \rightarrow \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}$ – with C_{dim} but without the compensating factor E_{char} , G would be too small by a factor of $E_{\text{char}} = 28.8$. Likewise: $\xi^2/(4m_e)$ has $[E^{-1}]$, not $[E^{-2}]$.
- Lepton mass table: $0.507/103.5/1815 \rightarrow 0.505/104.96/1783.4$ MeV (corpus ladder).
- (2 Oct. 2026) Notation f : field profile $f(\vec{r}, t)$ — not a frequency; note in the text (Doc. 190, R147).

Abstract

The T0 model describes physical phenomena through a universal energy field $E_{\text{field}}(x, t)$ with the parameter $\xi = \frac{4}{3} \times 10^{-4}$. The field equation is $\square E_{\text{field}} = 0$, the Lagrangian density $\mathcal{L} = \xi(\partial E)^2$. The model uses standard natural units with $\hbar = c = 1$.

Fundamental quantities:

- Characteristic energy: $E_0 = \sqrt{m_e \cdot m_\mu} = 7.348$ MeV (bare); fractally corrected $E_0/\sqrt{K_{\text{frak}}} = 7.398$ MeV (anchor for α)
- Fine structure constant: $\alpha = \xi(E_0/1 \text{ MeV})^2 \approx 1/137$
- Gravitational constant: $G = \xi^2/(4m_e) \times \text{factors}$

Predictions: Lepton masses with 2% accuracy, fine structure constant $\alpha \approx 1/137$ (from the masses alone $1/137.059$, +0.017 %).

Detailed derivations: See Document 011 (fine structure), 012 (gravitation), 018 (g-2 geometric), 019 (Lagrangian).

.1 Units Convention**T0 natural units (Heaviside-Lorentz)**

T0 uses Heaviside-Lorentz natural units:

$$\hbar = c = 4\pi\epsilon_0 = 1 \quad (1)$$

In this system:

- Fine structure constant: $\alpha = e^2 = 1$ (by convention in T0)
- Energy = Mass: $E = m$

- Length = Time = Energy⁻¹: $[L] = [T] = [E^{-1}]$

Note: In SI units, $\alpha \approx 1/137$. The choice $\alpha = 1$ in T0 is a unit convention that simplifies formulas.

Dimensions in natural units

$$[E] = E \quad (2)$$

$$[m] = E \quad (3)$$

$$[t] = E^{-1} \quad (4)$$

$$[L] = E^{-1} \quad (5)$$

$$[G] = E^{-2} \quad (6)$$

$$[\partial_\mu] = E \quad (7)$$

.2 Time-Energy Duality

Fundamental relation

$$T_{\text{field}}(x, t) \cdot E_{\text{field}}(x, t) = 1 \quad (8)$$

with $[T_{\text{field}}] = E^{-1}$ and $[E_{\text{field}}] = E$.

Intrinsic time field

$$T_{\text{field}}(x, t) = \frac{1}{E_{\text{field}}(x, t)} \quad (9)$$

.3 Universal Field Equation

Wave equation

$$\square E_{\text{field}} = 0 \quad (10)$$

with d'Alembert operator:

$$\square = \nabla^2 - \frac{\partial^2}{\partial t^2} \quad (11)$$

With sources

$$\nabla^2 E_{\text{field}} = 4\pi G\rho \cdot E_{\text{field}} \quad (12)$$

Dimension check: $[E^3] = [E^{-2}][E^4][E] = [E^3] \checkmark$

.4 Lagrangian Density

Universal Lagrangian density

$$\mathcal{L} = \xi \cdot (\partial_\mu E_{\text{field}})(\partial^\mu E_{\text{field}}) \quad (13)$$

with $\xi = \frac{4}{3} \times 10^{-4}$.

Euler-Lagrange equation

$$\frac{\partial \mathcal{L}}{\partial E} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu E)} = 0 \quad (14)$$

yields:

$$\square E_{\text{field}} = 0 \quad (15)$$

.5 Characteristic Energy

Definition

The characteristic energy E_0 is the geometric mean of electron and muon mass (derivation in Document 011):

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (16)$$

Numerical values

From experimental masses:

$$E_0 = \sqrt{0.511 \times 105.66} \quad (17)$$

$$= \sqrt{53.99} \quad (18)$$

$$= 7.348 \text{ MeV} \quad (19)$$

Fractally corrected value (Doc. 190, R72):

$$E_0^{T0} = \frac{E_0}{\sqrt{K_{\text{frak}}}} = \frac{7.348}{\sqrt{1 - 100\xi}} = 7.397 \approx 7.398 \text{ MeV} \quad (20)$$

Both values are the same E_0 , bare and fractally corrected: absolute values carry K_{frak} , ratios do not. The square-root factor follows because E_0 enters $\alpha = \xi E_0^2$ quadratically. The corrected value 7.398 MeV serves as the anchor for α .

(Corrected on 30 Sept. 2026: previously "Theoretical T0 value: $E_0^{T0} = 7.398 \text{ MeV}$; Deviation: 0.7 % (within geometric corrections)". The relation $7.398/7.348 = 1.00682 \approx 1/\sqrt{K_{\text{frak}}} = 1.00673$ (0.008 %) was not stated.)

Usage

E_0 serves as energy scale for:

- Fine structure constant: $\alpha = \xi (E_0/1 \text{ MeV})^2$
- Normalization of electromagnetic effects

.6 The Parameter ξ

Definition

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333 \times 10^{-4} \quad (21)$$

Dimensionless: $[\xi] = 1$.

Geometric components

$$\xi = G_3 \times S_{\text{ratio}} \quad (22)$$

where:

- $G_3 = \frac{4}{3}$: Geometric factor (sphere-cube ratio)
- $S_{\text{ratio}} = 10^{-4}$: Scale ratio

.7 Fine Structure Constant

In T0 units

In T0 natural units: $\alpha = 1$ (by convention)

Reconstruction of SI value

Important: The SI value can be reconstructed from T0 parameters!

$$\alpha_{\text{SI}} = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2 \quad (23)$$

Numerical calculation

With $\xi = \frac{4}{3} \times 10^{-4}$ and $E_0 = 7.398 \text{ MeV}$:

$$\alpha_{\text{SI}} = 1.3333 \times 10^{-4} \times (7.398)^2 \quad (24)$$

$$= 1.3333 \times 10^{-4} \times 54.73 \quad (25)$$

$$= 7.297 \times 10^{-3} \quad (26)$$

$$= \frac{1}{137.04} \quad (27)$$

Experimental: $\alpha_{\text{exp}} = \frac{1}{137.036}$

Agreement: 0.0005 % (with the anchor $E_0 = 7.398 \text{ MeV}$). From the masses alone, $\alpha = \xi E_0^2 / K_{\text{frak}}$ with $E_0 = 7.348 \text{ MeV}$ gives $\alpha = 1/137.059$ (+0.017 %).

(Corrected on 30 Sept. 2026: previously "Agreement: 0.03 %"; 1/137.035 against 1/137.036 is 0.0005 %.)

Dimension check

$$[\alpha_{\text{SI}}] = [\xi] \times \left[\frac{E}{E} \right]^2 = 1 \times 1 = 1 \quad \checkmark \quad (28)$$

.8 Gravitational Constant

T0 formula

The gravitational constant is derived from ξ and m_e (derivation in Document 012):

$$G = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (29)$$

(Corrected on 30 Sept. 2026: previously $\frac{\xi^2}{4m_e} \times C_{dim} \times C_{conv} - C_{dim} = 1/E_{char} = 3.472 \times 10^{-2}$ is cancelled again by the factor $E_{char} = 28.8$ in the step chain of Doc. 012; in the formula alone the product would give 2.35×10^{-12} instead of 6.674×10^{-11} .)

where:

- C_{dim} : Dimension correction
- C_{conv} : SI conversion factor

Fundamental relation

In natural units:

$$\xi = 2\sqrt{G \cdot m_e} \quad (30)$$

Solved for G :

$$G_{nat} = \frac{\xi^2}{4m_e} \quad (31)$$

Dimension: $[G] = [E^{-2}]$ in natural units; the right-hand side $\xi^2/(4m_e)$, however, has $[E^{-1}]$ – the missing factor $1/E_{char}$ and the SI conversion reside in C_{conv} (Doc. 012). (Corrected on 30 Sept. 2026: previously without this remark, so that $\xi^2/(4m_e)$ appeared as $[E^{-2}]$.)

.9 Characteristic Lengths

T0 characteristic length

$$r_0 = 2GE \quad (32)$$

Dimension: $[r_0] = [E^{-2}][E] = [E^{-1}] = [L] \checkmark$

Derivation

For spherically symmetric point source $\rho(r) = E_0 \delta^3(\vec{r})$:

Solution of $\nabla^2 E = 4\pi G \rho E$:

$$E(r) = E_0 \left(1 - \frac{r_0}{r}\right) \quad (33)$$

with $r_0 = 2GE_0$.

Time scale

$$t_0 = \frac{r_0}{c} = r_0 = 2GE \quad (34)$$

(since $c = 1$)

.10 Scale Hierarchy

Planck length as reference

$$\ell_P = \sqrt{G} = 1 \quad (\text{in nat. units}) \quad (35)$$

Scale ratio

$$\xi_{\text{ratio}} = \frac{\ell_P}{r_0} = \frac{\sqrt{G}}{2GE} = \frac{1}{2\sqrt{G} \cdot E} \quad (36)$$

For $E \sim 1$ GeV:

$$\frac{r_0}{\ell_P} = 2\sqrt{G}E = \frac{2E}{E_P} \sim 1.6 \times 10^{-19} \quad (\text{sub-Planck}) \quad (37)$$

(Corrected on 29 Sept. 2026: previously $r_0/\ell_P \sim 10^7$ – with $r_0 = 2GE$ and $\ell_P = \sqrt{G}$ one has $r_0/\ell_P = 2E/E_P$, i.e. 1.6×10^{-19} for $E = 1$ GeV.)

.11 Particles as Field Excitations

Classification by energy

Particle	Energy [MeV]
Electron	0.511
Muon	105.658
Tau	1776.86

Antiparticles

Negative field excitations: $E_{\text{field}} < 0$

.12 Lepton Masses

The T0 model predicts lepton masses (derivation in Document 003):

Lepton	T0 [MeV]	Exp [MeV]	Δ [%]
Electron	0.505	0.511	-1.18
Muon	104.96	105.66	-0.66
Tau	1783.4	1776.9	+0.37

(Corrected on 30 Sept. 2026: previously 0.507/103.5/1815 MeV with 0.87/2.09/2.16 % – the corpus ladder $m_i = r_i \xi^{P_i} v$ (Docs. 005, 006) gives 0.505/104.96/1783.4 MeV, i.e. -1.18/ -0.66/ +0.37 %; also $(0.511 - 0.507)/0.511$ would be 0.78 %, not 0.87 %.)

.13 Anomalous Magnetic Moments

Anomalous magnetic moments are treated in dedicated documents: Doc. 018 (with its revisions) and Doc. 158.

(Corrected on 30 Sept. 2026: previously this section gave the short formula $a_\ell = \frac{\xi}{2\pi}(E_\ell/E_e)^2$ with $a_\mu = 1.166 \times 10^{-3}$, $a_e = 2.122 \times 10^{-5}$ and $a_\tau = 1.28 \times 10^{-3}$. Recalculated, the formula gives $a_\mu = 2.122 \times 10^{-5} \times 42\,753 = 0.907$ and $a_\tau = 256.6$; a_e lies a factor of 55 below the measured value 1.15965×10^{-3} . The formula has been removed, as have its repetitions in the dimensional analysis, formula reference and calculation examples.)

.14 Three Field Geometries

Type 1: Localized spherical

$$E(r) = E_0 \left(1 - \frac{\beta}{r}\right), \quad \beta = r_0 \quad (38)$$

Application: Individual particles (electron, muon, tau)

Type 2: Localized non-spherical

$$E(\vec{r}) = E_0 \left(1 - \frac{\beta_{ij} r_i r_j}{r^3}\right) \quad (39)$$

Application: Bound systems

Type 3: Extended homogeneous

Effective parameter:

$$\xi_{\text{eff}} = \frac{\xi}{2} = \frac{2}{3} \times 10^{-4} \quad (40)$$

Application: Cosmology (see Document 026)

.15 Mathematical Identities

Energy field normalization

$$E_{\text{field}}(\vec{r}, t) = E_0 \cdot f(\vec{r}, t) \cdot e^{i\phi(\vec{r}, t)} \quad (41)$$

(Note of 2 Oct. 2026 on notation: here $f(\vec{r}, t)$ is the normalised space-time profile of the energy field, not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)
with:

- E_0 : Characteristic energy
- $f(\vec{r}, t)$: Normalized profile
- $\phi(\vec{r}, t)$: Phase

Duality consistency

Time-mass (Document 003): $T \cdot m = 1$

Time-energy (this document): $T \cdot E = 1$

In natural units ($c = 1$):

$$E = mc^2 = m \quad \Rightarrow \quad T \cdot m = T \cdot E \quad (42)$$

.16 Dimensional Analysis Verifications

Field equation

$$[\nabla^2 E] = [L^{-2}][E] = [E^2][E] = [E^3] \quad (43)$$

$$[4\pi G \rho E] = [E^{-2}][E^4][E] = [E^3] \quad \checkmark \quad (44)$$

Characteristic length

$$[r_0] = [2GE] = [E^{-2}][E] = [E^{-1}] = [L] \quad \checkmark \quad (45)$$

Lagrangian density

$$[\mathcal{L}] = [\xi][(\partial E)^2] = [1][E^4] = [E^4] \quad (\text{correct for Lagrangian density}) \quad (46)$$

(Corrected on 30 Sept. 2026: previously $[1][E^2] = [E^2]$; $[\partial_\mu E] = E^2$, hence $[(\partial E)^2] = E^4$, as in the symbol table.)

.17 Formula Reference

Fundamental equations

$$\text{Duality: } T_{\text{field}} \cdot E_{\text{field}} = 1 \quad (47)$$

$$\text{Wave equation: } \square E_{\text{field}} = 0 \quad (48)$$

$$\text{With sources: } \nabla^2 E = 4\pi G \rho E \quad (49)$$

$$\text{Lagrangian density: } \mathcal{L} = \xi(\partial E)^2 \quad (50)$$

Derived constants

$$\text{Characteristic energy: } E_0 = \sqrt{m_e \cdot m_\mu} = 7.348 \text{ MeV (bare; corrected 7.398)} \quad (51)$$

$$\text{Fine structure constant: } \alpha = \xi(E_0/1 \text{ MeV})^2 \approx 1/137 \quad (52)$$

$$\text{Gravitational constant: } G = \frac{\xi^2}{4m_e} \times \text{factors} \quad (53)$$

Characteristic scales

$$\text{T0 length: } r_0 = 2GE \quad (54)$$

$$\text{T0 time: } t_0 = 2GE \quad (55)$$

$$\text{Planck length: } \ell_P = \sqrt{G} = 1 \quad (56)$$

$$\text{Scale ratio: } \xi_{\text{ratio}} = \frac{1}{2\sqrt{GE}} \quad (57)$$

Prediction formulas

$$\text{Parameter: } \xi = \frac{4}{3} \times 10^{-4} \quad (58)$$

$$\text{Effective parameter: } \xi_{\text{eff}} = \frac{\xi}{2} \quad (59)$$

.18 Numerical Values

Fundamental constants (in natural units)

$$\hbar = 1 \quad (60)$$

$$c = 1 \quad (61)$$

$$\alpha = \frac{1}{137.036} \approx 7.297 \times 10^{-3} \quad (62)$$

$$G = 1 \text{ (numerically, dimension } [E^{-2}]) \quad (63)$$

T0 parameters

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333 \times 10^{-4} \quad (64)$$

$$\xi^2 = 1.7778 \times 10^{-8} \quad (65)$$

$$\frac{\xi}{2\pi} = 2.1221 \times 10^{-5} \quad (66)$$

$$\xi_{\text{eff}} = 6.6667 \times 10^{-5} \quad (67)$$

$$E_0 = 7.348 \text{ MeV (from exp. masses)} \quad (68)$$

$$E_0^{\text{T0}} = 7.398 \text{ MeV (fractally corrected, } E_0/\sqrt{K_{\text{frak}}}) \quad (69)$$

Lepton energies

$$E_e = 0.511 \text{ MeV} \quad (70)$$

$$E_\mu = 105.658 \text{ MeV} \quad (71)$$

$$E_\tau = 1776.86 \text{ MeV} \quad (72)$$

Energy ratios

$$\frac{E_\mu}{E_e} = 206.768 \quad (73)$$

$$\frac{E_\tau}{E_e} = 3477.2 \quad (74)$$

$$\frac{E_\tau}{E_\mu} = 16.817 \quad (75)$$

.19 Calculation Examples

Fine structure constant

Given:

- $\xi = 1.3333 \times 10^{-4}$
- $E_0 = 7.398 \text{ MeV}$

Calculation:

$$\left(\frac{E_0}{1 \text{ MeV}}\right)^2 = (7.398)^2 = 54.73 \quad (76)$$

$$\alpha = 1.3333 \times 10^{-4} \times 54.73 \quad (77)$$

$$= 7.297 \times 10^{-3} \quad (78)$$

$$= \frac{1}{137.04} \quad (79)$$

Experimental: $\alpha_{\text{exp}} = \frac{1}{137.036}$

Deviation: 0.0005 % (corrected on 30 Sept. 2026: previously 0.03 %)

Characteristic length (electron)

Given:

- $E_e = 0.511 \text{ MeV} = 0.511 \times 1.6 \times 10^{-13} \text{ J} = 8.2 \times 10^{-14} \text{ J}$
- $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
- $c = 3 \times 10^8 \text{ m/s}$

Conversion to SI units:

$$r_0 = \frac{2GE}{c^4} \approx 1.35 \times 10^{-57} \text{ m} \quad (80)$$

Planck comparison:

$$\frac{r_0}{\ell_P} = \frac{1.35 \times 10^{-57}}{1.6 \times 10^{-35}} \approx 8.4 \times 10^{-23} \quad (81)$$

(Corrected on 29 Sept. 2026: previously $r_0 = 2GE \approx 10^{-28} \text{ m}$ and $r_0/\ell_P \approx 10^7$ – with the SI values given, the formula reads $r_0 = 2GE/c^4$ and yields $1.35 \times 10^{-57} \text{ m}$; r_0 thus lies far below the Planck length.)

0.1 Symbol Reference

Symbol	Meaning	Dimension
ξ	Fundamental parameter	1
E_0	Characteristic energy	E
E_{field}	Universal energy field	E
T_{field}	Intrinsic time field	E^{-1}
r_0	T0 characteristic length	$L = E^{-1}$
t_0	T0 characteristic time	$T = E^{-1}$
ℓ_P	Planck length	$L = E^{-1}$
G	Gravitational constant	E^{-2}
α	Fine structure constant	1
E_e, E_μ, E_τ	Lepton energies	E
m_e, m_μ, m_τ	Lepton masses (= E in nat. units)	E
$\mathcal{L}$	Lagrangian density	E^4
$\square$	d'Alembert operator	E^2
ξ_{eff}	Effective parameter ($\xi/2$)	1

0.2 Unit Conversions

Natural → SI

$$1 \text{ (Energy)} = 1 \text{ GeV} = 1.6 \times 10^{-10} \text{ J} \quad (82)$$

$$1 \text{ (Length)} = \frac{\hbar c}{1 \text{ GeV}} = 0.197 \text{ fm} \quad (83)$$

$$1 \text{ (Time)} = \frac{\hbar}{1 \text{ GeV}} = 6.58 \times 10^{-25} \text{ s} \quad (84)$$

Standard natural units

In standard convention ($\hbar = c = 1$):

- $\alpha = \frac{e^2}{4\pi\epsilon_0} \approx \frac{1}{137}$ (dimensionless)
- All quantities in powers of energy
- Physical predictions identical to other conventions

0.3 Relations to Other Documents

- **Document 003:** Time-mass duality, foundations, origin of ξ
- **Document 011:** Fine structure constant (geometric derivation)
- **Document 012:** Gravitational constant (systematic derivation)
- **Document 018:** Geometric g-2 formulation (fractal geometry)
- **Document 019:** Lagrangian formulation (quantum field theory)
- **Document 026:** Cosmology ($\xi_{\text{eff}} = \xi/2$)
All formulations are based on $\xi = \frac{4}{3} \times 10^{-4}$.